GreenKeeper: A Budget Friendly Open-Source Autonomous Lawn-mower

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***Abstract*—This paper presents the design and development of GreenKeeper, a budget-friendly autonomous lawn-mower, aimed at providing high-end performance within a budget-friendly price range of $500 to $1000. Existing budget autonomous lawn-mowers typically rely on physical wire boundaries, which significantly increase the overall cost and setup complexity. GreenKeeper eliminates the need for such boundaries while ensuring reliable navigation and obstacle avoidance. Key features include a GPS-based boundary system, powered by an RTK-GPS module offering centimeter-level accuracy, and a computer vision system for dynamic obstacle detection using YOLOv8n and MiDaS. GreenKeeper's movement is managed through a simple pathing algorithm integrated with an inertial measurement unit (IMU), while power is supplied by a 25V lithium iron phosphate (LiFePO4) battery. The design balances cost, functionality, and performance, demonstrating the potential for high-quality autonomous lawn care within an affordable price range.**

***Index Terms*—Autonomous Navigation, Computer Vision, Control Systems, GPS, Sensors, User Interface.**

I. Introduction

A recent survey shows that 94% of Americans care deeply about the appearance of their lawn [1]. With world temperatures on the rise each year, it has become increasingly more dangerous for individuals to be able to tackle lawn-care safely [2]. Moreover, elderly individuals and those with disabilities are often unable to work, or are far more restricted in the work they can do in lawn maintenance. As a result, these groups are often locked out of being able to address their lawn in a safe, easy, and affordable manner.

Within the autonomous lawn-mower industry, there exists a budget-friendly range, where prices range between $500 and $1000. Devices such as the WORX Landroid S 1/8 Acre [3] and the YARDCARE E400 [4], can be found within this price range. The Landroid S 1/8 Acre utilizes an above ground boundary wire set up manually by the user, installed with stakes placed into the ground. Autonomous lawn-mowers in this price range require the user to set up a physical wire boundary above ground, or pay extra for an underground wire installation to keep the device within a certain perimeter. Underground boundary wires are typically not removed or further utilized upon transferring of residence, leading to waste. Above ground wires require manual set up, which some users may be unable to accomplish by themselves. These existing budget options meet their design goals, but the additional cost, setup complexity, and waste of a boundary wire, either below or above ground, leaves much to be desired.

After the budget-friendly range is the high-end price bracket, where prices range between $1000 to around $5000, but some are as much as $19,350, like the L400i Deluxe Ambrogio [5]. In this range, autonomous lawn-mowers begin to perform with very satisfactory results while typically not requiring the installation of a physical boundary wire. Models like the Navimow i105 [6] ($999.99) and the Dreame A1 [7] ($1699.99) can be purchased in the lower bracket of this price range. The Navimow i105 utilizes AI-assisted lawn mapping as well as obstacle detection, and the Dreame A1 utilizes similar technologies for a 3D omnidirectional obstacle avoidance system along with a wireless boundary setup. Autonomous lawn-mowers in this price range typically use a combination of cameras and sensors like LIDAR to navigate their way around a lawn while avoiding obstacles. LIDAR is great, however it is also exceedingly expensive. Some of the higher end models like the Husqvarna Automower 430XH [8] ($1749.99) still use a physical boundary wire that most cheaper models rely on. The existing options in this price range are no longer an easy or budget option for the average American household, and some even still rely on wasteful outdated technology.

This paper presents GreenKeeper, a prototype for an autonomous lawn-mower capable of maximizing the features seen in higher end models, while removing the need for a physical boundary wire via the use of RTK-GPS technology and drastically cutting costs to $430-$450 a unit.

The rest of this paper is organized as follows. The design, including the mechanical and electrical components of GreenKeeper are presented in Section II. Section III presents the software solutions utilized in GreenKeeper. The experimental results are presented in Section IV. Current limitations and future work are presented in Section V, and finally the conclusions are provided in Section VI.

II. Hardware Implementation

This section presents the hardware implementation of GreenKeeper. A block diagram of the components of GreenKeeper is presented in Fig. 1.

A. Mechanical Components

1) Chassis/Body: The chassis is the physical body to which all other components and sensors are fixed to [9]. It was formed from plywood sheets via laser cutting. This simple design allowed to more quickly test and prototype the boundary, pathing, and obstacle avoidance systems.

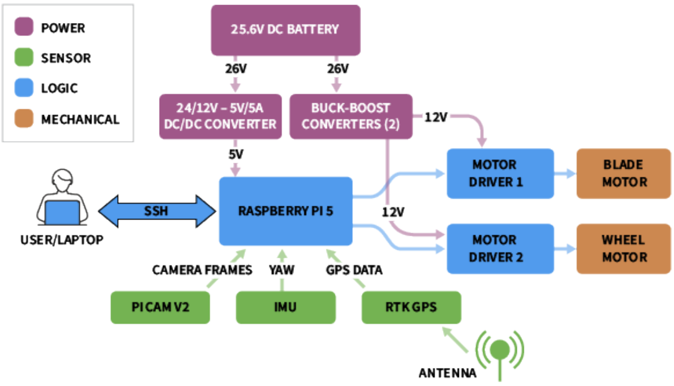

Fig. 1. A block diagram of hardware components. The GPS, IMU, and Motors are the primary sensors and control mechanisms. Not pictured in this figure is the active cooling module, which sits directly on top of the Pi 5's CPU.

2) Blade Mechanism & Wheels: The blade mechanism is 3D-printed in PLA filament. Grooves in the outer perimeter allow for the blades to be screwed securely into place. The blade mechanism connects underneath the chassis to the large brushless DC (BLDC) motor. The rear driving wheels are printed in the same filament as the blade mechanism and are spiked to aid in the handling of rough terrain and steep slopes.

B. Electrical Components

1) Raspberry Pi 5: A 4GB Raspberry Pi 5 [10] was used as the primary and only processing unit. The Pi 5 contains all the necessary pins to connect all modules and other controllers, such as the two motor drivers, the Pi camera, RTK-GPS module, and IMU. The Raspberry Pi 5 was chosen for its ability to handle lite models of computer vision software.

2) CPU Cooling: The Pi 5 Active Cooling Unit [11] was utilized to keep the Pi from overheating, especially during vision model inference. It is a simple heatsink with fan that sits on top of the Pi 5's CPU, and connects directly to a dedicated pin connector on the Pi 5 motherboard.

3) Camera: The PiCamera Module v2 [12] is mounted on a plywood sheet at the front of the chassis. Camera feed is utilized in video format for both vision models, as well as in remote-control (RC) mode for the users viewing.

4) RTK-GPS (and Antenna): A 40-Pin RTK-GPS module connecting directly to the Pi 5 was utilized along with an Antenna to receive dual band signals from both L1 and L5 satellites. This is deployed in both the mapping and pathing functionalities.

5) DC Motors: Three 12-volt DC motors are used in the design; two brushed DC motors are used to propel and steer GreenKeeper from the rear, and one BLDC motor is mounted in the center, with the axle poking down under the chassis to connect the blade mechanism.

6) Motor Drivers: Two H-Bridge motor drivers [13] [14] were utilized to control and supply current to the 3 motors.

7) 26.5v DC Battery: A 26.5v DC Lithium Iron Phosphate (LiFePO4) battery [15] is utilized to power the entire device, including the Pi 5, components, and motors.

8) Power Converters: A 24v-5v/5A DC/DC converter was utilized to meet the Pi 5's rather unconventional power requirements (5v/5A). To limit voltage and provide current to the 12v motors, two variable buck/boost converters were utilized to connect the battery to each of the motor drivers respectively.

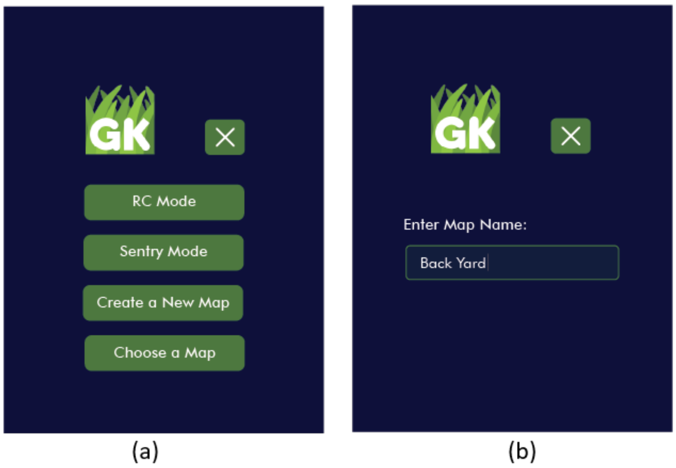

Fig. 2. Examples from the app used to control and execute commands on GreenKeeper. (a) A view of the main menu. (b) User prompt to enter map name after selecting "create a new map".

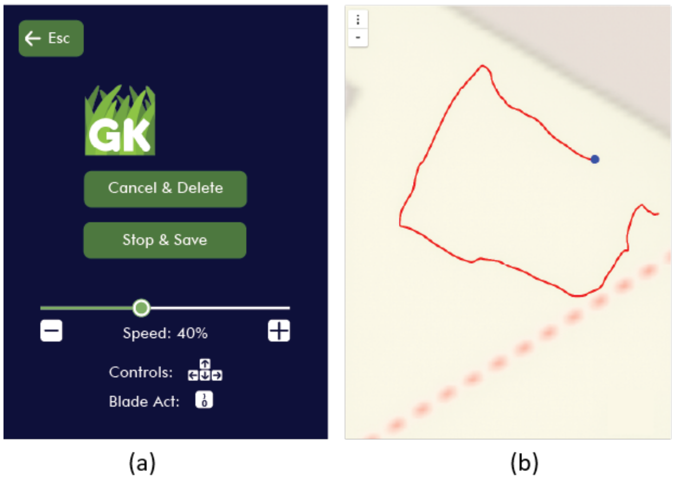

Fig. 3. Examples of the mapping mode within the app. (a) Mapping mode state screen with RC controls. (b) A corresponding live GPS map output.

9) Inertial Measurement Unit: The inertial measurement unit (IMU) utilized was the MPU6050 [16]. This module was used to track yaw to be able to make accurate and precise turns while pathing.

III. Software Implementation

A. User Interface

To enable user control, the PyGame library was utilized to display a graphical user interface (GUI) alongside a simple state machine to control GreenKeeper for each menu item. Currently available states include: main menu, RC mode, sentry mode, mapping mode, and pathing mode.

RC mode allows the user to control GreenKeeper's movement and actuate the blade mechanism with keyboard controls while viewing the camera feed (without vision models), allowing for a low latency remote control experience with lawn-

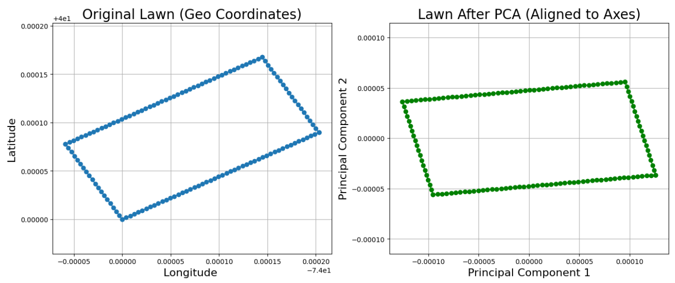

Fig. 4. Comparison of lawn shape before and after PCA alignment. The left plot (blue) shows the slanted rectangular plotted using latitude and longitude coordinates. The same shape is transformed using PCA which approximately aligns the major axis of the shape with the horizontal axis (Principal Component 1).

mowing capabilities. Sentry mode allows the user to view the camera feed along with detected objects from the vision models while GreenKeeper remains stationary. Mapping mode prompts the user for a map name, then activates the RTK-GPS module and begins creating a GPS point map as the user guides GreenKeeper around a desired lawn perimeter . Lastly, pathing mode allows the user to select a previously generated GPS map, then activates the pathing algorithm enabling GreenKeeper to automatically plan and traverse a route covering the map.

B. Lawn Mapping

To create effective lawn maps, an RTK-enabled GPS module was used. This module provides an accuracy of 1-5cm on average. To map out a lawn, the user is first prompted to enter a name for the lawn, as seen in Fig. 2(a), which creates an empty text file for writing GPS coordinates in the Pi 5's directory. Then the user will control GreenKeeper and guide it around the perimeter of the lawn they would like to set a boundary for. GPS points are saved to the text file as the user traverses the perimeter. These points are later converted into a polygon and utilized in the pathing algorithm as the boundary for GreenKeeper.

C. Obstacle Detection

To perform object detection, two pre-trained vision models are utilized along with the camera feed from the PiCam Module v2. The first model is YOLOv8n [17] which has been converted to ncnn format, the fastest and smallest version of YOLOv8 object detection for ARM architecture edge devices, like the Pi 5. YOLOv8n detects objects and annotates the frame with bounding boxes and confidence ratings. The resulting image is passed through a second model called MiDaS [18] [19] from Intel Labs, which estimates the distance of the object with respect to the camera.

To increase efficiency and save power, these obstacle detection processes are currently only called during the sentry mode operation. It is also important to note that unlike many typical vision tasks, the most important metric for GreenKeeper is the recall (the model's ability to detect an object exists in the first place) rather than the accuracy or classification.

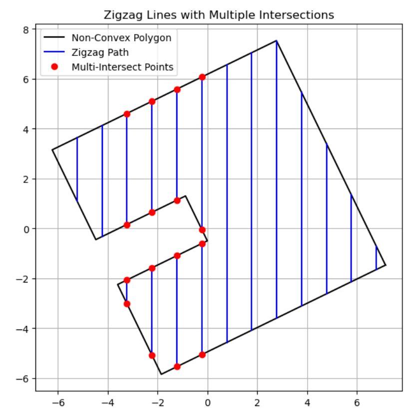

Fig. 5. Complete lawn map with predetermined zig zag path that GreenKeeper will follow. Note that the map shown in Fig 5 is different from the maps displayed in Fig 4.

D. Pathing Algorithm

To tackle automated pathing, a multi-threaded pathing program was developed that enables GreenKeeper to autonomously traverse a lawn using its GPS, IMU, and motor drivers. This program runs independently of other modules like computer vision or the main menu, and is composed of three main components: map loading, path generation, and real-time path following with visualization.

The program is structured as a class that initializes the motor and IMU drivers. The IMU relies on its gyroscope sensor to find its yaw value ranging from 0° to 360°. When initializing the IMU, an initial calibration is performed by grabbing the first 100 samples of the gyroscope, and find the average yaw rate. That value is then subtracted from any new samples to reduce the amount of drift the robot experiences. After initializing the drivers, the program loads a saved GPS point map, and generates a zigzag path using the Shapely library [20] to define the lawn boundary as a polygon. The IMU is used for precise rotation between path points.

Once inside the lawn boundary, GreenKeeper applies Principal Component Analysis (PCA) [21] to align the polygon with the primary axes, simplifying path generation. Zigzag lines are spaced 0.5 meters apart, which is narrower than GreenKeeper's width of 1.5ft. Starting from the bottom-left, vertical lines are generated by connecting (x, miny) to (x, maxy) within bounds. Lines alternate direction until the rightmost edge is reached.

Finally, the zigzag path is transformed back to the original GPS coordinate system using PCA's inverse transform and stored for execution, as seen in Fig 5.

IV. Results

A. Final Prototype

The final prototype of GreenKeeper is presented in Fig. 6. The spiked 3-D printed rear wheels along with the weight of the battery give good traction on a variety of terrains and slopes. Unlocked castor wheels at the front allow for smooth low friction turns.

Not pictured in Fig. 6 is the blade mechanism, which attaches onto the large central BLDC Motor. The 3D CAD

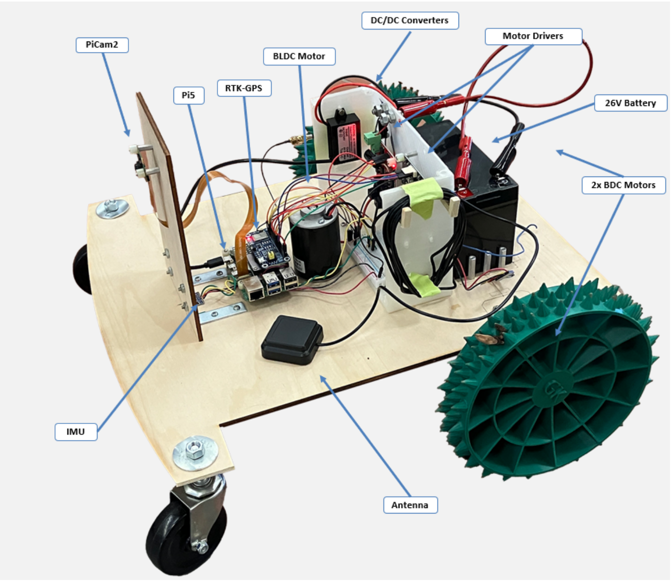

Fig. 6. Final prototype of GreenKeeper.

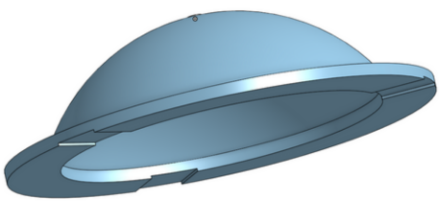

Fig. 7. The blade mechanism which attaches underneath the chassis to the central DC motor. Note the 3 grooves in the lower outer perimeter, this is where the blades are screwed into place. This mechanism attaches to the central BLDC Motor, which rotates between 1000 and 3000 RPM

TABLE I
IMU Accuracy Evaluation

Expected Angle (°)	Measured Average (°)	Absolute Error (°)	Accuracy (%)
0	0.30	0.30	99.83
90	89.42	0.58	99.68
180	176.68	3.32	98.15

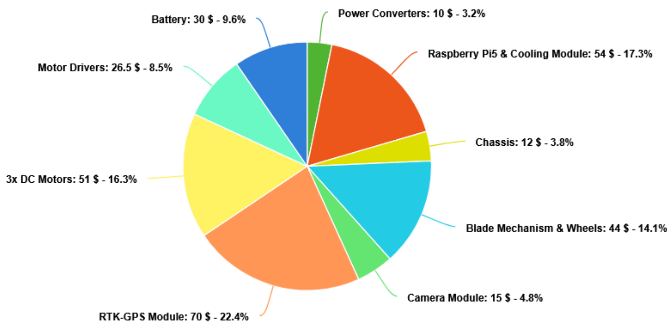

Fig. 8. Cost breakdown of the initial prototype

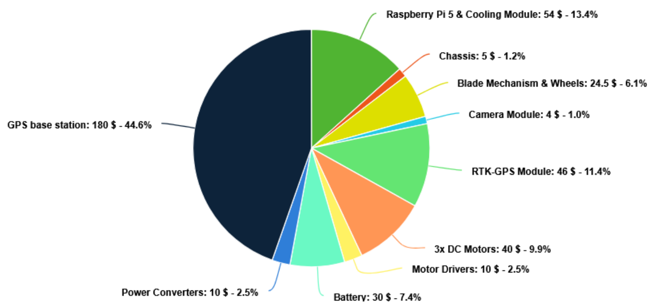

Fig. 9. Cost breakdown of mass producing 1000+ units utilizing an injection molded chassis and including the addition of a GPS base station.

diagram for the blade mechanism is shown in Fig. 7. It was printed in the same color & material as the rear driving wheels.

B. Inertial Measurement Unit (MPU6050)

GreenKeeper uses the MPU-6050 gyroscope to estimate orientation by measuring angular velocity and integrating it over time to compute yaw. This process introduces minor drift due to integration error, even when stationary. During testing, the IMU was calibrated to 0° at startup and manually rotated to test accuracy at various orientations. Each angle/orientation was tested by averaging 10 readings. Although manual rotation introduced slight user error, accuracy remained above 98% as seen in Table 1, which is sufficient. Since GreenKeeper primarily moves in straight lines, small orientation errors do not significantly affect performance.

C. Cost of Production

The total cost of the prototype was $430, where the cost breakdown for the prototype is shown in Fig. 8. Fig. 9 presents the cost breakdown of producing 1000+ units of Greenkeeper including the addition of a GPS base station [22] as discussed in the next section.

The bulk ordering of the hardware components as well as using an injection molded chassis rather than a laser-cut plywood one would lower production costs. Considering these factors, it would be feasible to develop a functionally identical product to this $430 prototype, for about $444, while increasing the reliability of the pathing function, and resistance to hard weather conditions. Notice that the cost would be much less than the available boundary wire free models available, such as the Husqvarna Automower 430XH ($1749.99) [8], or the Dreame A1 ($1699.99) [7]. Also note that this cost is expected to increase due to marketing, storage, transportation, and development of the device. Table II provides a comparison of GreenKeeper with other autonomous lawn-mowers in terms of available features and price.

V. Design Limitations & Future Work

There are several areas where the design of GreenKeeper could be improved. Currently, the latency introduced by the GPS communication creates a risk of pathing malfunction. With the addition of an on-site GPS base station [22], the pathing program could become far more accurate and responsive. As discussed in the cost analysis, an injection molded

TABLE II
COMPARISON OF ROBOTIC LAWN MOWER SPECIFICATIONS.

Feature	**WR165 Landroid S**	**Navimow i105N**	**Dreame Robotic-mower A1**	**Automower 430XH**	**GreenKeeper**
Price	$699	$999.00	$1,599.99	$1,749.99	$500
Area Capacity	0.5 Acre (~2000 m^2)	0.125 Acre	0.5 Acre (~2000 m^2)	0.8 Acre (~3200 m^2)	0.25 Acre (1000 m^2)
Cutting Width	180 mm	180 mm	218 mm	240 mm	200 mm
Cutting Height	30–60 mm	51–91 mm	30–70 mm	50–90 mm	60 mm
Maximum Slope	30% / 17°	29% / 16°	45% / 24°	15% (boundary) – 45% / 8.5° – 24°	N/A
Noise Level	59 dB	58 dB	N/A	58 dB	N/A
Boundary Type	Physical Wire	Wireless	Wireless	Physical Wire	Wireless
Battery Capacity (mAh)	6000	2550	5000	5000	6000
Charging Type	Charging Dock	Charging Dock	Charging Dock	Charging Dock	Plug-in
Sensors	- Ultrasonic - Boundary Sensor - Rain - IMU	- RTK-GPS - Camera(s) - Odometer - IMU	- Lidar - Rain - IMU	- GPS	- Camera - RTK-GPS - IMU

chassis would provide further weather resistance, as well as greatly lower overall costs, which would make up for the addition of a GPS base station.

Further research on the system should also include analyzing design changes to the power delivery system which can greatly improve the efficiency and cost of the system. A specially developed power conversion custom PCB integrated for all components would further drive down costs, and substantially increase the efficiency of the device.

The auto-pathing state should also be expanded with the obstacle detection system. Algorithms such as potential field avoidance [23] or A* pathing [24] could be utilized to find a new path around detected objects. This could be further extended by using a Spiral Spanning Tree coverage path planner [25] [26] that allows GreenKeeper to have any starting point within the perimeter.

Sentry mode will relay any detected objects along with their estimated distance back to the users device over SSH, but currently does not prompt any further actions to be taken by GreenKeeper. This could be complimented with an optional notification system.

There are also a number of avenues to improve the user experience and GUI. The current GUI, while completely functional as a prototype, could be updated to be more visually appealing and user-friendly. Utilizing technology like Bluetooth Low-Energy (BLE) alongside a multi-platform mobile phone app would allow some of the workload to be taken off the Pi 5. It would also allow for a cleaner user interface and connection experience, as well as a more reliable mapping feature that would not require a WiFi connection.

VI. CONCLUSIONS

This paper presented GreenKeeper, an accessible and affordable prototype of an autonomous lawn-mower capable of competing with higher end models while remaining within the budget-friendly range. GreenKeeper achieves this by utilizing RTK-GPS mapping to replace the typical need for a physical boundary wire and provide a feedback source for the mower's pathing to follow. The integration of computer vision models like YOLOv8n and MiDaS show strong promise for real time object detection & distance estimation on affordable edge devices such as the Pi 5. Compared to existing products that rely on expensive sensors like LIDAR, GreenKeeper is able to provide similar functionalities for a fraction of the price.

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